

problem becomes

$$\left\{ \begin{array}{ll} \dot{u}_i(t) - \nabla \cdot (\alpha(t) \nabla u_i(t)) + (\nabla \cdot \mathbf{w}(t) + \beta(t)) u_i(t) = f_i(t) & \text{in } \Omega_i(t), \\ u_i(t) = 0 & \text{on } \partial\Omega_i(t) \setminus \Gamma(t) \\ & \text{for } i = 1, 2, \\ u_1(t) = u_2(t) & \text{on } \Gamma(t), \\ \alpha(t) \nabla u_1(t) \cdot \nu_1(t) + \alpha(t) \nabla u_2(t) \cdot \nu_2(t) = 0 & \text{on } \Gamma(t), \end{array} \right. \quad (2)$$

where $\nu_i(t)$ is the unit outward normal vector of $\partial\Omega_i(t)$, $f_i(t) = f(t)|_{\Omega_i(t)}$, and $u_i(t) = u(t)|_{\Omega_i(t)}$.

The non-overlapping domain decompositions can then be derived by approximating the transmission problem. For example, consider the classic Robin–Robin method, first introduced in [27]. By taking linear combinations of the last two equations in (2), one has the equivalent Robin conditions

$$\alpha(t) \nabla u_1(t) \cdot \nu_i(t) + s_0 u_1(t) = \alpha(t) \nabla u_2(t) \cdot \nu_i(t) + s_0 u_2(t) \quad \text{on } \Gamma(t) \text{ for } i = 1, 2,$$

and a method parameter $s_0 > 0$. Alternating between the subdomains then gives the Robin–Robin method as computing (u_1^n, u_2^n) for $n = 1, 2, \dots$ with

$$\left\{ \begin{array}{ll} \dot{u}_1^n(t) - \nabla \cdot (\alpha(t) \nabla u_1^n(t)) + (\nabla \cdot \mathbf{w}(t) + \beta(t)) u_1^n(t) = f_1(t) & \text{in } \Omega_1(t), \\ u_1^n(t) = 0 & \text{on } \partial\Omega_1(t) \setminus \Gamma(t), \\ \alpha(t) \nabla u_1^n(t) \cdot \nu_1(t) + s_0 u_1^n(t) = & \\ \alpha(t) \nabla u_2^{n-1}(t) \cdot \nu_1(t) + s_0 u_2^{n-1}(t) & \text{on } \Gamma(t), \\ \dot{u}_2^n(t) - \nabla \cdot (\alpha(t) \nabla u_2^n(t)) + (\nabla \cdot \mathbf{w}(t) + \beta(t)) u_2^n(t) = f_2(t) & \text{in } \Omega_2(t), \\ u_2^n(t) = 0 & \text{on } \partial\Omega_2(t) \setminus \Gamma(t), \\ \alpha(t) \nabla u_2^n(t) \cdot \nu_2(t) + s_0 u_2^n(t) = & \\ \alpha(t) \nabla u_1^n(t) \cdot \nu_2(t) + s_0 u_1^n(t) & \text{on } \Gamma(t). \end{array} \right. \quad (3)$$

Here, u_2^0 is an initial guess and $u_i^n(t)$ approximates $u_i(t) = u(t)|_{\Omega_i(t)}$. Note that the Robin–Robin method is sequential, but the computation of each u_i^n can be implemented in parallel when $\Omega_i(t)$ is a union of nonadjacent subdomains, as is the case in Figure 1.

The well posedness of parabolic equations on evolving domains can be derived via the framework [3], which relies on a variational formulation where the standard Sobolev–Bochner solution space

$$H^1((0, T); H^{-1}(\Omega)) \cap L^2((0, T); H_0^1(\Omega)) \quad (4)$$

is generalized to evolving domains $\Omega(t)$. The framework has also been extended to a Banach space setting [2]. This variational setting constitutes the starting point of the design and analysis of a wide range of finite element methods for equations on evolving domains. The